Independent Model Validation Report - Home Credit PD Scorecard

Scope

Validation of the WOE logistic scorecard (champion) vs LightGBM (challenger) on a held-out test set of 61,503 applications (default rate 8.07%), covering discrimination, calibration, stability, benchmarking, sensitivity and model assumptions in the spirit of SR 11-7.

Results

Category Test Value Threshold Status
Discrimination Test AUC 0.7465 >=0.70 PASS, >=0.65 CONDITIONAL PASS
Discrimination Test Gini 0.4929 >=0.40 PASS, >=0.30 CONDITIONAL PASS
Discrimination Test KS 0.3697 >=0.30 PASS, >=0.20 CONDITIONAL PASS
Calibration Hosmer-Lemeshow p-value 0.0933 >0.05 PASS, >0.01 CONDITIONAL PASS
Calibration Brier skill score 0.0739 >0.05 PASS, >0 CONDITIONAL PASS
Stability Score PSI train vs test 0.0001 <0.10 PASS, <0.25 CONDITIONAL PASS
Stability Features with CSI > 0.1 0 0 PASS, <=2 CONDITIONAL PASS
Benchmark Challenger minus scorecard AUC 0.0193 p>=0.05 PASS, else CONDITIONAL CONDITIONAL
Sensitivity Max mean PD change for +/-10% shift (%) 9.02 <=10 PASS, <=25 CONDITIONAL PASS
Assumptions Max VIF 1.77 <=5 PASS, <=10 CONDITIONAL PASS
Assumptions WOE monotonic features 21/21 all PASS, else CONDITIONAL PASS

Overall rating: Satisfactory with findings

11 tests: 10 PASS, 1 CONDITIONAL, 0 FAIL.

Calibration by decile

Decile N Predicted rate Actual rate
1 6,151 0.0133 0.0138
2 6,150 0.0229 0.0211
3 6,150 0.0314 0.0265
4 6,150 0.0405 0.0431
5 6,151 0.0512 0.0481
6 6,150 0.0644 0.0605
7 6,150 0.0815 0.0846
8 6,150 0.1053 0.1050
9 6,150 0.1437 0.1509
10 6,151 0.2492 0.2536

Sensitivity (mean PD change, %)

Feature -10% +10%
EXT_SOURCE_3 6.56 -5.64
EXT_SOURCE_2 9.02 -8.51
EXT_SOURCE_1 1.94 -1.73
DAYS_EMPLOYED 0.7 -0.67
PREV_REFUSED_RATE -0.21 0.69

Findings and recommendations

- CONDITIONAL: Challenger minus scorecard AUC = 0.0193 (threshold p>=0.05 PASS, else CONDITIONAL).
- Re-run this validation on an out-of-time sample once later vintages are available.
- Review EXT_SOURCE dependence and run a fairness assessment before production use.